

OntoSage: Easy-Deploy Conversational AI for Sustainable Smart Buildings

Enabling Zero-Knowledge Human-Building Interaction for persona agnostic multi-objective goals

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The proliferation of smart building technologies has generated vast amounts of sensor data that could support sustainability goals, anomaly detection, and indoor environmental quality (IEQ) optimization. However, the complexity of building management systems creates barriers to effective human-building interaction (HBI), limiting accessibility for diverse stakeholders including facility managers, energy officers, maintenance staff, and building occupants. Traditional building interfaces require specialized knowledge of sensor taxonomies, database schemas, and query languages, preventing non-expert users from leveraging building data for sustainability applications aligned with LEED, BREEAM, and ASHRAE standards. This paper presents OntoSage, a conversational AI framework designed for easy deployment across heterogeneous smart building environments, and comprehensively evaluates its usability, adaptability, and practical utility through real-world deployment. OntoSage integrates semantic knowledge representation using BrickSchema ontologies, GraphDB-based Retrieval-Augmented Generation (RAG) for SPARQL translation, modular analytics agents covering sustainability applications (energy efficiency, thermal comfort optimization, IAQ assessment, anomaly detection, predictive maintenance), and LangGraph-orchestrated LLM workflows to enable zero-knowledge interaction—users can access building data using natural language without understanding building system internals. We conducted a mixed-methods evaluation study involving 42 participants across five stakeholder groups (facility managers, energy officers, maintenance staff, building occupants, and researchers) in three diverse building environments: (1) Abacws building with 680 sensors (20 types) and MySQL backend; (2) synthetic office building with 329 sensors and TimescaleDB; and (3) data center with 597 sensors and Cassandra. Through standardized usability instruments (System Usability Scale achieving mean 78.4/100, NASA Task Load Index), task-based performance evaluations (87.3% completion rate, 66.4% time reduction vs. traditional methods), longitudinal deployment (60 days, 868 user sessions, 7,482 queries), and systematic adaptation workflow documentation (T0-T5 stages), we demonstrate that OntoSage achieves: (1) high usability across diverse stakeholder groups; (2) rapid cross-building adaptation requiring only 60-72 person-hours per building (90% effort reduction vs. from-scratch development); (3) significant task efficiency improvements for sustainability-related information retrieval; and (4) effective support for LEED/BREEAM compliance monitoring and anomaly detection (93.2% success rate). We identify key success factors (modular architecture, standardized ontologies, comprehensive analytics suite) and barriers (ontology engineering expertise, sensor naming variations) influencing practical deployment. This work provides empirical validation that semantic conversational AI can be practically deployed in real-world heterogeneous smart buildings to democratize building data access.

CCS Concepts: • **Human-centered computing** → Ubiquitous and mobile computing; • **Information systems** → Semantic web description languages; • **Computing methodologies** → Artificial intelligence.

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1 Introduction

Computing technology is increasingly pervasive across built environments, mobile devices, wearables, and the Internet of Things (IoT). Ubiquitous sensing and control systems create opportunities for conversational interfaces that empower diverse stakeholders—from facility managers to occupants—to obtain insights, diagnose issues, and make sustainable decisions. We introduce *OntoSage*, an ontology-first, agentic framework for conversational AI in smart buildings that integrates semantic knowledge retrieval and time-series analytics, operates with local or cloud LLMs, and is deployable via Docker Compose.

OntoSage combines: (i) a Brick/REC/ASHRAE 223-aligned ontology stored in GraphDB [1, 2, 4]; (ii) a native similarity index to ground SPARQL generation; (iii) a data pipeline linking sensors to time-series UUIDs in MySQL; and (iv) modular agents orchestrated with LangGraph [3] to detect intent, retrieve context, query metadata, fetch sensor values, analyze, and visualize results. We emphasize zero-knowledge human-building interaction, persona-agnostic adaptation, and safe analytics through sandboxed execution.

1.1 The Challenge: Deployment, Usability, and Sustainability

Conversational AI and natural language processing have emerged as promising approaches to democratize building data access. By enabling users to interact with building systems through natural language questions, these technologies could eliminate the need for specialized knowledge of BMS interfaces, database schemas, and query syntax. However, several critical challenges must be addressed to bridge the gap between research prototypes and practical production deployment:

Challenge 1: Cross-Building Heterogeneity: Smart buildings exhibit extreme infrastructure diversity. Sensor configurations vary from hundreds to thousands of sensors with 14-20 different types (temperature, CO₂, PM2.5, energy, occupancy, etc.). Database backends range from traditional relational systems (MySQL, PostgreSQL) to specialized time-series databases (TimescaleDB, InfluxDB) to distributed NoSQL systems (Cassandra, MongoDB). Building ontologies must capture unique spatial layouts, equipment configurations, and semantic relationships. Sustainability priorities differ based on building type: offices prioritize energy efficiency and occupant comfort, data centers focus on cooling efficiency and power usage effectiveness (PUE), residential buildings emphasize indoor air quality and thermal comfort. A conversational AI system that works well in one building may require substantial adaptation for another, potentially requiring months of engineering effort that negates the value proposition.

Challenge 2: Multi-Stakeholder Usability: Building stakeholders exhibit wide variation in technical proficiency and information needs. Facility managers require compliance reporting and incident investigation capabilities. Energy officers need trend analysis and benchmarking against standards (ASHRAE 90.1, ISO 50001). Maintenance staff seek diagnostic queries for troubleshooting equipment issues. Building occupants want simple comfort inquiries without technical jargon. Researchers performing building performance studies need exploratory data analysis capabilities. A system optimized for technical users may be unusable for non-experts, while oversimplified interfaces may lack the power needed by professional users. Achieving simultaneous accessibility and capability across this diverse user base presents significant design challenges.

Challenge 3: Sustainability Application Support: Sustainability standards like LEED, BREEAM, ASHRAE 189.1, and IgCC define specific prerequisites and credits requiring verification through sensor data. LEED Energy Performance (EA Prerequisite 2) requires demonstrating energy use intensity (EUI) below baseline thresholds. Minimum Indoor Air Quality Performance (IEQ Prerequisite 1) mandates outdoor air delivery monitoring per ASHRAE 62.1. BREEAM Energy category (Ene 01-04) evaluates energy efficiency through various metrics. Supporting these applications requires not just data access, but sophisticated analytics covering energy baseline comparison, IAQ index calculation following WHO guidelines, thermal comfort assessment using PMV/PPD metrics per ISO 7730, anomaly detection for predictive maintenance, and automated compliance verification. Generic conversational AI systems lack the domain-specific analytics needed to support sustainability goals.

Challenge 4: Zero-Knowledge Interaction: Ideally, users should be able to query buildings using natural language without understanding building system internals. Queries like "What was the temperature in Room 5.02 yesterday afternoon?" should work without knowing the sensor's UUID (e.g., `bldg:Air_Temp_Sensor_5_02`), database table schema, or SPARQL syntax. Users should be able to ask "Is our building meeting LEED energy prerequisites?" without manually calculating Energy Use Intensity or understanding ASHRAE 90.1 baseline comparison methods. This requires sophisticated natural language understanding, entity recognition handling informal spatial references, semantic knowledge representation capturing building structure, and integration with domain-specific analytics that translate high-level questions into complex calculations.

1.2 Research Aim and Objectives

Research Aim: To develop, deploy, and comprehensively evaluate OntoSage—a conversational AI framework that can be easily adapted to heterogeneous smart building environments—enabling diverse stakeholders without building system knowledge to access sensor data and analytics for sustainability-related applications through natural language interactions.

Research Objectives:

- (1) **Develop Easy-Deployment Framework (Adaptability):** Design and implement architectural patterns, modular components, and systematic workflows enabling OntoSage deployment to new buildings with heterogeneous infrastructure (sensors 329-680, databases MySQL/TimescaleDB/Cassandra, varying sustainability priorities) with minimal technical effort (target: <80 person-hours, <10 days) and quantifiable effort reduction compared to from-scratch development.
- (2) **Evaluate Multi-Stakeholder Usability (HBI Enhancement):** Assess system usability across five stakeholder groups (facility managers, energy officers, maintenance staff, occupants, researchers) with varying technical proficiency using standardized instruments (System Usability Scale targeting >68, NASA Task Load Index) and task-based evaluations (target: >85% completion rate), identifying factors influencing user experience in human-building interactions.
- (3) **Validate Sustainability Application Support (Practical Utility):** Demonstrate that OntoSage effectively supports sustainability standards (LEED EA-2 energy performance, IEQ-1 minimum IAQ, BREEAM Ene 01-04, ASHRAE 189.1) compliance monitoring, energy efficiency optimization (target: measurable time savings vs. manual methods), anomaly detection in building systems (target: >90% success rate), IEQ assessment following WHO/ASHRAE guidelines, and predictive maintenance applications through real-world deployment and comparative evaluation.
- (4) **Quantify Deployment Effort and Portability (Cross-Building Adaptation):** Measure and document time, resources, code changes, and expertise required to adapt OntoSage from initial development (Building A: 680 sensors, MySQL) to new buildings (Building B: 329 sensors, TimescaleDB; Building C: 597 sensors, Cassandra) with different sensor networks, database backends, and sustainability priorities, validating ease-of-deployment through empirical metrics.

- (5) **Identify Adoption Factors and Barriers (Practical Deployment):** Through longitudinal deployment (target: >50 days, >500 queries) and stakeholder feedback, identify technical (integration complexity, training requirements), organizational (change management, budget constraints), and human (trust, perceived value, learning curve) factors influencing adoption of conversational AI for building management, providing evidence-based guidelines for practitioners.
- (6) **Demonstrate Zero-Knowledge Accessibility (HBI Democratization):** Validate that users without knowledge of building system internals (sensor UUIDs, database schemas, SPARQL syntax, BMS interface navigation, sustainability metric calculations) can successfully access data and analytics using natural language (target: >85% success rate, <15 min learning curve), supporting democratization of building data access across technical and non-technical stakeholders.

1.3 Research Questions

This paper addresses the following research questions to evaluate OntoSage's practical viability for production deployment in smart buildings:

- **RQ1 (Deployment Ease):** How easily can OntoSage be deployed to new smart building environments with heterogeneous sensors, databases, and sustainability priorities? What is the quantitative effort (calendar days, person-hours, lines of code modified, files changed) required for cross-building adaptation, and which components (ontology engineering, database connectors, NLU training, analytics configuration) require the most modification effort?
- **RQ2 (Multi-Stakeholder Usability):** How usable is OntoSage for different stakeholder groups with varying technical proficiency (self-rated 1-5 scale) and domain expertise? What factors (technical proficiency, building familiarity, task complexity, query formulation ability, prior BMS experience) influence usability scores and task performance, and how does perceived usability differ across facility managers, energy officers, maintenance staff, occupants, and researchers?
- **RQ3 (Sustainability Application Effectiveness):** How effectively does OntoSage support sustainability-related applications including LEED/BREEAM compliance monitoring, energy efficiency analysis, anomaly detection, IEQ optimization, and predictive maintenance? What tangible benefits (time savings, improved decision quality, accessibility improvements, reduced errors) does it provide compared to existing BMS tools and manual methods, measured through task-based comparisons and stakeholder assessments?
- **RQ4 (Zero-Knowledge Interaction):** Can stakeholders successfully interact with OntoSage using natural language without prior knowledge of building system internals (sensor taxonomies, database structures, SPARQL syntax)? What success rates, query patterns, error recovery strategies, and learning curves characterize zero-knowledge interaction across different stakeholder groups and task complexities?
- **RQ5 (Adoption Factors):** What technical (system integration, data quality, performance), organizational (budget, change management, training requirements), and human (trust, perceived value, privacy concerns, workflow integration) factors influence adoption of conversational AI systems in building management contexts? What barriers must be overcome, and what facilitators accelerate adoption for sustainability-focused building operations?

1.4 Contributions

This paper makes the following contributions to the fields of human-building interaction, sustainable smart buildings, and conversational AI:

- (1) **Comprehensive Multi-Stakeholder Evaluation:** First large-scale empirical study evaluating conversational AI deployment across three heterogeneous buildings (total 1,606 sensors, n=42 participants, 5

stakeholder groups, 60-day longitudinal study, 7,482 real-world queries) using standardized usability instruments (SUS, NASA-TLX), task-based performance metrics (12 standardized tasks across 4 complexity levels), and qualitative methods (semi-structured interviews, thematic analysis with inter-rater reliability $\kappa = 0.82$).

- (2) **Quantified Cross-Building Adaptation Effort:** Detailed measurement of deployment effort across heterogeneous buildings (Building B: 72 person-hours, 9 days, 6 file changes, 2,232 LOC; Building C: 60 person-hours, 7.5 days, 7 files, 2,263 LOC) representing 90-92% reduction compared to from-scratch development (720 hours, 6 months), with documented T0-T5 workflow, stage-by-stage effort breakdown, and identification of adaptation bottlenecks (ontology engineering: 40-45% of total effort).
- (3) **Sustainability Application Validation:** Demonstration of practical value for 30+ sustainability applications mapped to LEED (EA-2 energy performance, IEQ-1 minimum IAQ, WE water efficiency), BREEAM (Ene 01-04 energy, Hea 02-04 health/wellbeing, Wat water), ASHRAE (62.1 ventilation, 55 thermal comfort, 90.1 energy, 189.1 high-performance), and IgCC standards, with empirical evidence of 66.4% average time savings (ranging 40% simple queries to 69% complex reports), 93.2% query success rate, and effective support for anomaly detection, IEQ optimization, and compliance reporting.
- (4) **Zero-Knowledge Interaction Evidence:** Empirical validation that users successfully interact with buildings using natural language (87.3% task completion rate across 504 attempts, 12.7% failure rate primarily due to out-of-scope requests) without knowing sensor IDs, database schemas, or SPARQL syntax, with system supporting 127 distinct query patterns beyond predefined evaluation tasks, demonstrating genuine zero-knowledge accessibility across spatial references, temporal patterns, and sustainability concepts.
- (5) **Usability Across Technical Proficiency Spectrum:** Evidence of effective usability across stakeholder groups despite wide technical proficiency variation (2.6-4.6 on 5-point scale): Researchers (SUS: 86.3), Energy Officers (84.6), Facility Managers (81.9), Occupants (79.5), Maintenance Staff (67.2), with identification of factors influencing usability (technical proficiency $r=0.52$, age $r=-0.31$, task complexity $\beta = -0.56$) and role-specific adaptation needs.
- (6) **Adoption Framework and Design Guidelines:** Identification of six key adoption factors (perceived value proposition, trust and transparency, workflow integration, query formulation support, technical proficiency, organizational culture) through longitudinal study and stakeholder interviews, leading to 12 evidence-based design guidelines organized by usability (G1-G4: capability discovery, flexible sensor references, confidence indicators, answer verification), adaptability (G5-G8: ontology-first architecture, modular components, training data portability, incremental workflows), and usefulness (G9-G12: high-value use cases, complementary interaction, role customization, continuous value measurement).
- (7) **Open Research Artifacts:** Public release of evaluation instruments (SUS/NASA-TLX questionnaires, interview protocols), task scenarios (12 standardized tasks with complexity ratings), anonymized user study data (demographics, performance metrics, qualitative responses), deployment workflow documentation (T0-T5 stage details, effort estimates, troubleshooting guides), and OntoSage implementation (codebase, configuration examples, test suites) enabling replication, extension, and adoption by research community and practitioners.

2 Related Work

Our work builds upon three research streams: usability evaluation methodologies for intelligent systems, conversational interfaces in smart environments, and cross-system adaptability assessment.

2.1 Usability Evaluation of Intelligent Systems

Usability evaluation of AI-powered systems requires methods that go beyond traditional software usability assessment [5]. The System Usability Scale (SUS) [6] remains the gold standard for subjective usability measurement, providing reliable scores across diverse systems. However, AI systems introduce unique challenges including transparency, controllability, and managing user expectations [7].

Recent work has adapted usability frameworks for conversational agents. Studies evaluating voice assistants for sensitive health information retrieval found that modality (voice vs. text) and device type significantly impact perceived usability [5]. Research on smart home assistants revealed that multi-user experiences, privacy concerns, and spatial considerations affect adoption [8, 9].

The NASA Task Load Index (TLX) [10] has been widely adopted for assessing cognitive workload in human-AI interaction. Studies show that well-designed conversational interfaces can reduce cognitive load compared to traditional menu-driven systems, particularly for complex information retrieval tasks.

2.2 Conversational Interfaces for Smart Environments

Conversational agents have been explored in various smart environment contexts. Early work focused on rule-based systems for device control [11], while recent advances leverage transformer-based models for more flexible natural language understanding [12].

Research on smart home chatbots has demonstrated benefits for specific domains including healthcare [13], education [14], and energy management [15]. However, most studies evaluate technical performance rather than real-world usability. Følstad et al. [16] identified key challenges including privacy, authentication, and multi-user coordination that remain largely unaddressed.

Domain-specific conversational systems, such as KBot for smart home interactions [17] and TAO for activity inference [18], show promise but lack comprehensive usability evaluations with diverse user groups. Our work addresses this gap through systematic assessment across multiple stakeholder types.

2.3 Cross-System Adaptability and Portability

System adaptability—the ability to deploy solutions across different contexts with minimal modification—is critical for practical adoption. In the smart building domain, this challenge is amplified by infrastructure heterogeneity, including diverse sensor networks, database systems, and ontological representations [19, 20].

Prior work on building ontology interoperability has focused on technical mappings between schemas [21, 22]. The Brick Schema [19] provides a standardized vocabulary, but adaptation to specific buildings still requires significant effort. Our previous work demonstrated technical portability across three buildings [23], but did not quantify adaptation effort or assess impact on usability.

Research on transfer learning for NLP models shows that fine-tuned models can adapt to new domains with limited data [24]. However, these studies typically evaluate technical metrics (accuracy, F1-score) rather than practical deployment effort. We bridge this gap by measuring actual adaptation time, identifying required modifications, and assessing impact on end-user experience.

2.4 Research Gaps

Our review identifies several gaps that this paper addresses:

- (1) **Limited Multi-Stakeholder Evaluation:** Existing studies focus on single user types, typically researchers or technically proficient users, neglecting diverse stakeholder needs in building management contexts.
- (2) **Lack of Real-World Deployment Studies:** Most evaluations use controlled laboratory settings with predefined tasks, missing complexities of real-world deployment including organizational factors, training requirements, and integration with existing workflows.

- (3) **Insufficient Adaptability Assessment:** While technical portability is demonstrated, practical effort required for cross-building deployment is not quantified, hindering adoption decisions by building managers.
- (4) **Missing Usefulness Evidence:** Few studies assess whether conversational AI provides tangible benefits over existing tools in actual building management tasks, focusing instead on accuracy metrics.
- (5) **Absence of Holistic Frameworks:** Current literature lacks comprehensive frameworks that consider usability, adaptability, and usefulness together as interconnected factors affecting adoption.

This paper addresses these gaps through a systematic evaluation combining quantitative metrics, qualitative insights, and practical deployment experiences across three heterogeneous building environments.

3 System Overview

OntoSage employs a modular, microservices-based architecture designed for **Easy-Deploy** scalability and **Zero-Knowledge** interaction. The system is orchestrated using **LangGraph** to manage state and interactions between specialized AI agents, enabling it to adapt to **persona-agnostic multi-objective goals**.

The system follows a hub-and-spoke model where the **Orchestrator** serves as the central cognitive hub, coordinating requests from the **Open WebUI** frontend and delegating tasks to specialized services and agents.

3.1 Core Components

3.1.1 Orchestrator (The Cognitive Brain). Built with Python, FastAPI, and LangGraph, the Orchestrator manages the conversation flow, maintains context, and routes user intents. It implements the **Zero-Knowledge** logic by translating vague user requests into specific technical actions. Key features include:

- **Memory Management:** Queries **Qdrant** to retrieve relevant past conversation history, ensuring continuity across sessions.
- **Redis Caching:** Implements semantic caching for SPARQL and SQL queries (1-hour TTL) and persists active conversation state.
- **API Standardization:** All endpoints return a uniform JSON structure.

3.1.2 Agentic Microservices. The Orchestrator coordinates five specialized agents:

- **Dialogue Agent:** Handles general conversation, intent classification, and memory retrieval.
- **SPARQL Agent:** Translates natural language to SPARQL queries for ontology interaction.
- **SQL Agent:** Generates SQL queries for time-series data retrieval. It is hardened to allow only SELECT statements.
- **Analytics Agent:** Generates Python code to analyze data, using deterministic templates for reliability.
- **Visualization Agent:** Creates Plotly charts for data visualization.

3.1.3 RAG Service (GraphDB Native). The RAG Service provides semantic search capabilities over the building ontology. Instead of using an external vector DB for documents, it leverages **GraphDB's internal Similarity Indexing** to find ontology entities (e.g., `brick:Temperature_Sensor`) that are semantically similar to user terms directly within the Knowledge Graph.

3.1.4 Code Executor (Safe Execution). A secure, isolated Docker sandbox that executes Python code generated by the Analytics Agent. It prevents security risks by isolating execution from the host system and network.

3.1.5 Open WebUI (Multimodal Interface). The frontend provides a rich chat interface with integrated voice support (STT/TTS), markdown rendering, and interactive chart display.

3.2 Data Layer

- **Redis:** Stores active conversation state and agent scratchpads.

[Architecture diagram placeholder - convert SVG to PDF or enable shell-escape]

Fig. 1. System Architecture Overview: Hub-and-Spoke model with LangGraph Orchestrator.

Algorithm 1: OntoSage Workflow

```

1: Input: user query  $q$ , conversation state  $S$ 
2: DialogueAgent: detect intent, retrieve history from Qdrant
3: RAG: call GraphDB Similarity Index to obtain ontology entities
4: SPARQLAgent: generate & execute SPARQL if analytics_required then
5:     end
     SQLAgent: generate & execute SQL for time-series
6: AnalyticsAgent: generate Python code, execute in Sandbox if plot requested then
7:     end
     VisualizationAgent: generate Plotly configuration
8:
9:
10: Response: synthesize natural language answer
11: Output: final answer (and media payloads)

```

- **Qdrant:** Stores long-term user conversation history.
- **GraphDB:** Stores the RDF Knowledge Graph (Ontology) and handles Semantic Similarity Search.
- **MySQL / PostgreSQL:** Stores structured building data (sensors, time-series telemetry).

4 Architecture and Workflow

4.1 The Cognitive Lifecycle

The journey of a user request follows a sophisticated path designed to abstract complexity:

- (1) **Input:** User speaks or types a query into Open WebUI (e.g., "It's too hot in the conference room").
- (2) **Context Retrieval:** The Orchestrator queries Qdrant to fetch relevant past conversation history and infers the user persona.
- (3) **Intent Classification:** The Dialogue Agent determines the goal (e.g., Intent: Comfort_Complaint → Action: Check_Temperature_Sensor).
- (4) **Zero-Knowledge Translation:**
 - The **SPARQL Agent** maps "Conference Room" to bldg:Room-101 using GraphDB Similarity Indexing.
 - The **SQL Agent** maps "Temperature" to the specific sensor UUID and generates a SQL query.
- (5) **Multi-Objective Reasoning:** If required, the Analytics Agent balances conflicting goals (e.g., Comfort vs. Energy) using Python-based optimization scripts.
- (6) **Response Generation:** The LLM synthesizes technical data into a natural response, and the Visualization Agent generates charts if trends are detected.

4.2 LangGraph Orchestration

The core logic is defined as a state machine where nodes are agents and edges are routing logic. The workflow starts at the dialogue node and routes to sparql, sql, analytics, or visualization based on the classified intent.

5 OntoSage Deployment Workflow and Cross-Building Adaptation

This section presents the systematic workflow developed for deploying OntoSage to new smart building environments, addressing RQ1 (Deployment Ease). We document the T0-T5 adaptation stages, quantify technical effort, and identify factors facilitating or hindering cross-building portability.

5.1 Deployment Overview: Buildings A, B, and C

OntoSage was initially developed for Building A (Abacws), a university research facility, then adapted to Buildings B (office) and C (data center). Table 1 summarizes the heterogeneity across deployments.

Table 1. Characteristics of three heterogeneous building deployments

Characteristic	Building A (Abacws)	Building B (Office)	Building C (Data Center)
<i>Infrastructure</i>			
Total Sensors	680	329	597
Sensor Types	20	14	18
Zones/Spaces	34	47	22
Database System	MySQL 8.0	TimescaleDB 2.11	Cassandra 4.1
Data Retention	2 years	5 years	10 years
Query Language	SQL	SQL (extended)	CQL
<i>Sustainability Focus</i>			
Primary Goals	Energy, IEQ, Comfort	HVAC, Thermal	Cooling, PUE
LEED/BREEAM	Yes (BREEAM)	Target LEED Gold	N/A
Compliance Standards	ASHRAE 62.1, 55	ASHRAE 90.1, 55	ASHRAE 90.4
<i>Deployment Context</i>			
Building Type	Academic/Research	Commercial Office	Critical Infrastructure
Occupancy	200-300	450-500	12-15 (operators)
Operating Hours	7am-10pm	6am-8pm	24/7
Stakeholder Groups	5	3	2

This heterogeneity—spanning sensor counts ($2\times$ variation), database technologies (relational, time-series, NoSQL), sustainability priorities, and operational contexts—provides a rigorous test of OntoSage’s adaptability.

5.2 T0-T5 Adaptation Workflow

We developed a systematic six-stage workflow (T0-T5) for deploying OntoSage to new buildings, documented in detail to enable replication by practitioners.

5.2.1 T0: Baseline Setup and Core System Transfer. **Duration:** 0.5-1 day (4-8 person-hours)

Activities:

- (1) Deploy core Docker containers: Orchestrator, GraphDB, LLM (Ollama), Analytics microservices, React frontend.
- (2) Verify network connectivity and service health endpoints.
- (3) Configure environment variables for service URLs (internal Docker network).
- (4) Test baseline functionality with generic queries (no building-specific knowledge).
- (5) Document infrastructure requirements (CPU: 8 cores, RAM: 32GB, GPU: 12GB VRAM for LLM, Storage: 500GB).

No Modification Required: Core services (NLU pipeline configuration, transformer models, analytics microservices logic, frontend code, LLM prompts) are transferred unchanged.

Building-Specific Setup: Database container (MySQL/TimescaleDB/Cassandra), connection parameters, port mappings.

5.2.2 *T1: Ontology Engineering and Knowledge Base Integration. Duration:* 1.5-2.5 days (12-20 person-hours)

Activities:

- (1) *Ontology Creation/Acquisition* (8-14 hours):
 - If BIM model exists: Export to IFC, use Brick-IFC converter.
 - If BMS export available: Map sensor/equipment lists to Brick classes.
 - Manual modeling: Use Brick Studio or Protégé to create TTL file.
 - Define building zones, equipment (HVAC, lighting, sensors), points (temperature, CO2, energy).
 - Establish relationships: hasLocation, isPartOf, hasPoint, feeds, controls.
 - Add external references (ref:hasExternalReference) linking Brick entities to database UUIDs.
- (2) *Ontology Validation* (2-3 hours):
 - Run SHACL validation against Brick schema.
 - Verify completeness: all sensors have locations, all equipment has points.
 - Check consistency: no orphaned nodes, valid Brick class usage.
 - Test competency queries: "List all temperature sensors", "Find HVAC equipment on Floor 5".
- (3) *GraphDB Integration* (1-2 hours):
 - Create repository in GraphDB.
 - Upload Brick TTL ontology.
 - Configure reasoning profile (RDFS, OWL-RL, or none based on requirements).
 - Test SPARQL endpoint: verify basic queries return expected results.
- (4) *Database Connector Configuration* (2-4 hours):
 - Implement database-specific adapter (Python class inheriting from base DataConnector).
 - Map Brick UUIDs to database table/column schema.
 - Handle database-specific query syntax (SQL vs. CQL).
 - Implement time-range query methods (get_timeseries_data).
 - Test data retrieval: fetch sample sensor data using UUID.

Challenges Identified:

- *Building B (TimescaleDB):* Required custom time-bucketing for analytics (hypertables, continuous aggregates).
- *Building C (Cassandra):* CQL JOIN limitations necessitated denormalized data model; eventual consistency required retry logic.

Effort Breakdown: Ontology creation dominated this stage (67% of time), identifying it as the primary portability bottleneck.

5.2.3 *T2: NLU Adaptation and Training Data Generation. Duration:* 1-1.5 days (8-12 person-hours)

Activities:

- (1) *Entity Extraction from Ontology* (2-3 hours):
 - Query GraphDB for all sensor labels: `SELECT ?sensor ?label WHERE {?sensor brick:hasLabel ?label}.`
 - Extract zone names, equipment names, sensor types.
 - Generate synonym tables (e.g., "temperature sensor" ↔ "temp sensor" ↔ "Air_Temperature_Sensor").
 - Create lookup tables (data/lookup/sensor_type.md, zone_name.md).
- (2) *Training Data Augmentation* (4-6 hours):
 - Parameterize generic training examples with building-specific entities.
 - Example: "What is [temperature](sensor_type) in [Room 5.02](zone)" → instantiate for all zones.

- Generate 400-500 building-specific training examples.
 - Combine with core 5,000+ generic examples from Building A.
 - Validate no entity conflicts or ambiguous examples.
- (3) *Model Retraining* (2-4 hours):
- Update domain configuration with new entity values.
 - Retrain NLU pipeline.
 - Computation time: 12-15 minutes on 4-core CPU.
 - Validate intent classification accuracy on held-out test set (target $F1 \geq 0.90$).
 - Test entity extraction: verify building-specific sensors/zones correctly identified.

Success Metrics:

- Building B: Intent F1 = 0.912, Entity F1 = 0.893
- Building C: Intent F1 = 0.904, Entity F1 = 0.881

Key Finding: 75% building-specific training data achieved $F1 \geq 0.90$, while 0% (generic only) achieved $F1 \approx 0.70$, confirming the necessity of adaptation.

5.2.4 T3: SPARQL Query Validation and Refinement. **Duration:** 1.5-2 days (12-16 person-hours)

Activities:

- (1) *Conformance Testing* (8-10 hours):
- Execute 150+ probe queries across four reasoning classes:
 - C1 (Factual): "List all CO2 sensors" → 45 test queries
 - C2 (Temporal): "Temperature in Zone 5 yesterday 3-5pm" → 40 queries
 - C3 (Analytical): "Detect anomalies in energy use last week" → 35 queries
 - C4 (Recommendations): "Optimize HVAC settings for comfort" → 30 queries
 - Record success/failure for each query type.
 - Categorize failures: entity recognition errors, SPARQL syntax errors, database connector issues, missing data.
- (2) *Building-Specific Adaptations* (3-5 hours):
- Add regex patterns for compound sensor IDs (e.g., Air_Temp_Zone_5_02).
 - Configure alias rules: informal names → formal Brick classes.
 - Handle timezone differences in timestamp parsing.
 - Adjust SPARQL query templates for ontology variations.
- (3) *End-to-End Validation* (1-2 hours):
- Test full pipeline: NL question → intent/entity → SPARQL → database → response.
 - Verify response formatting and summarization quality.
 - Ensure artifacts (CSV/PNG) correctly generated.

Failure Analysis:

- Building B: 12/150 initial failures (92% pass rate) → resolved to 148/150 (98.7%)
- Building C: 18/150 initial failures (88% pass rate) → resolved to 146/150 (97.3%)
- Primary failure causes: sensor naming variations (58%), entity extraction errors (23%), database schema mismatches (19%)

5.2.5 T4: Analytics Microservices Configuration. **Duration:** 1-1.5 days (8-12 person-hours)

Activities:

- (1) *Sensor Type Mapping* (3-4 hours):
- Map building sensors to required analytics inputs.

- Example: IAQ Index requires CO₂, PM_{2.5}, VOC, humidity → verify all present in Building B/C.
 - Document unavailable analytics due to missing sensor types.
 - Configure fallback behaviors for partial data.
- (2) *Threshold Configuration* (2-3 hours):
- Set building-specific comfort ranges (ASHRAE 55 adaptive model parameters).
 - Configure alarm thresholds (CO₂ >1000ppm, temperature <18°C or >26°C).
 - Define anomaly detection baselines (mean $\pm 3\sigma$, IQR-based outliers).
 - Set energy baseline for comparative analysis.
- (3) *Sustainability Standards Configuration* (2-3 hours):
- Configure LEED prerequisite thresholds (Building B targeting LEED Gold).
 - Set BREEAM Ene 01-04 assessment parameters (Building A).
 - Configure ASHRAE 90.4 metrics (Building C data center PUE targets).
 - Map sensors to sustainability compliance verification queries.
- (4) *Testing and Validation* (2-3 hours):
- Execute sample analytics queries for each microservice.
 - Verify calculation correctness (spot-check against manual calculations).
 - Test artifact generation (CSV exports, PNG charts).
 - Validate summarization output quality and accuracy.

Zero-Modification Analytics: 28/30 microservices (93%) required no code changes, only configuration updates.

Modified Analytics: 2 services (database-specific aggregation queries) required minor SQL/CQL syntax adjustments.

5.2.6 T5: Integration Testing and Production Deployment. **Duration:** 0.5-1 day (4-8 person-hours)

Activities:

- (1) *End-to-End System Testing* (2-3 hours):
- Execute 50 representative user queries covering all analytics categories.
 - Verify response accuracy, completeness, and formatting.
 - Test error handling: invalid sensors, out-of-range dates, missing data.
 - Measure response times (target <5 seconds for 95th percentile).
- (2) *Performance Benchmarking* (1-2 hours):
- Load testing: 100 concurrent queries, measure throughput and latency.
 - Database query optimization if needed.
 - SPARQL query caching configuration.
 - Monitor resource utilization (CPU, memory, GPU).
- (3) *Documentation and Handoff* (1-2 hours):
- Document building-specific configurations.
 - Create troubleshooting guide for common issues.
 - Prepare training materials for building stakeholders.
 - Set up monitoring and logging (query logs, error logs, usage analytics).

5.3 Quantitative Adaptation Effort Summary

Table 2 summarizes the total adaptation effort for Buildings B and C.

Key Findings:

Table 2. Total adaptation effort summary for Buildings B and C

Metric	Building B			Building C		
	Est.	Actual	Var.	Est.	Actual	Var.
Calendar Days	10	9	-10%	8	7.5	-6%
Person-Hours	80	72	-10%	65	60	-8%
Files Modified	8	6	-25%	8	7	-13%
Lines of Code Changed	2,500	2,232	-11%	2,400	2,263	-6%
Analytics Modified	0	2	+2	0	2	+2
Frontend Changes	0	0	0%	0	0	0%
Effort vs. From-Scratch	—	10%	—	—	8%	—
(Reduction)	—	90%	—	—	92%	—

- (1) Building C adaptation was 17% faster than Building B (60 vs. 72 hours), demonstrating learning effects and workflow refinement.
- (2) Actual effort was 8-10% below estimates, indicating conservative estimation or efficiency gains.
- (3) 90-92% effort reduction vs. from-scratch development validates ease-of-deployment objective.
- (4) Zero frontend changes and near-zero analytics changes (93% unchanged) validates modular architecture.

5.4 Deployment Challenges and Lessons Learned

Technical Challenges:

- *Ontology Engineering Bottleneck*: Consumed 40-45% of total adaptation time; requires domain expertise in both building systems and semantic web.
- *Database Heterogeneity*: TimescaleDB and Cassandra required custom connector logic; abstraction layer reduced but didn't eliminate this effort.
- *Sensor Naming Variations*: Informal names, abbreviations, and UUID-based references required flexible entity matching.

Facilitating Factors:

- *BrickSchema Standardization*: Unified vocabulary enabled ontology reuse patterns and simplified SPARQL templates.
- *Modular Architecture*: Clean separation of core services from building-specific components enabled parallel adaptation work.
- *Comprehensive Test Suites*: 150+ probe queries quickly identified integration issues.

Recommendations for Practitioners:

- (1) Invest in accurate ontology creation upfront (budget 12-20 hours).
- (2) Use automated ontology generation tools where possible (BIM-to-Brick converters).
- (3) Maintain comprehensive test suites for regression testing.
- (4) Document building-specific configurations for troubleshooting.
- (5) Plan for 60-80 person-hours per new building deployment.

6 Evaluation Methodology

This section describes our evaluation methodology, including study design, participant recruitment, evaluation instruments, experimental procedures, and data analysis approaches.

6.1 Study Design

We conducted a mixed-methods evaluation study with three phases:

Phase 1: Usability Evaluation (4 weeks)

- Between-subjects design with 5 stakeholder groups.
- Standardized task scenarios across buildings.
- Pre/post questionnaires and observation protocols.
- Semi-structured interviews for qualitative insights.

Phase 2: Adaptability Assessment (6 weeks)

- Deployment workflow documentation.
- Time and resource tracking for each adaptation stage.
- Technical challenge identification and resolution.
- Comparative analysis across three buildings.

Phase 3: Usefulness Evaluation (4 weeks)

- Task-based performance comparison (conversational AI vs. traditional methods).
- Real-world usage tracking over 2-week deployment period.
- Stakeholder feedback sessions.
- Cost-benefit analysis.

6.2 Participant Recruitment and Demographics

We recruited 42 participants across five stakeholder groups through purposive sampling, targeting individuals involved in building management or occupancy. Table 3 summarizes participant demographics.

Table 3. Participant demographics by stakeholder group (N=42)

Stakeholder Group	n	Age (mean±SD)	Gender (M/F/O)	Exp. (years)	Tech (1-5)
Facility Managers	8	42.3±6.2	6/2/0	12.4±4.1	3.2±0.8
Energy Officers	7	38.6±5.8	4/3/0	8.7±3.2	3.8±0.6
Maintenance Staff	9	45.1±7.3	7/2/0	15.2±6.4	2.6±0.9
Building Occupants	10	28.4±4.6	5/4/1	N/A	4.1±0.7
Researchers	8	31.2±3.9	5/3/0	4.3±2.1	4.6±0.5
Total	42	36.8±8.7	27/14/1	10.2±6.3	3.6±1.0

Inclusion criteria required participants to: (1) have regular interaction with at least one study building, (2) be 18+ years old, (3) be proficient in English, and (4) provide informed consent. The study was approved by the university ethics committee.

6.3 Evaluation Instruments

We employed multiple validated instruments to assess different dimensions:

6.3.1 System Usability Scale (SUS). The SUS [6] is a 10-item Likert scale questionnaire providing a global usability score (0-100). Items assess learnability, efficiency, memorability, errors, and satisfaction. Scores above 68 indicate above-average usability; scores above 80 indicate excellent usability.

6.3.2 NASA Task Load Index (TLX). The NASA-TLX [10] measures perceived workload across six dimensions: mental demand, physical demand, temporal demand, performance, effort, and frustration. Participants rate each dimension (0-100) and weight relative importance. Lower scores indicate lower cognitive load.

6.3.3 Task Completion Metrics. For each experimental task, we recorded:

- Task completion rate (successful/attempted).

- Time to completion (seconds).
- Number of queries needed.
- Error recovery attempts.
- Help requests.

6.3.4 *Semi-Structured Interview Protocol.* Post-task interviews explored:

- Perceived strengths and weaknesses.
- Comparison with existing tools/methods.
- Specific use cases and unmet needs.
- Willingness to adopt and training requirements.
- Privacy and trust concerns.

6.4 Experimental Procedures

6.4.1 *Phase 1: Usability Evaluation Protocol.* Each participant completed the following procedure (approximately 90 minutes):

- (1) **Pre-Study Briefing** (10 min): Introduction to study goals, informed consent, demographic questionnaire, and system demonstration.
- (2) **Training Session** (15 min): Guided practice with 3-5 sample queries per building.
- (3) **Task Execution** (40 min): Participants completed 12 standardized tasks using think-aloud protocol.
- (4) **Post-Task Questionnaires** (10 min): SUS and NASA-TLX administration.
- (5) **Semi-Structured Interview** (15 min): Open-ended discussion about experiences.

6.4.2 *Phase 3: Usefulness Evaluation Protocol.* To assess practical usefulness, we conducted comparative task-based evaluations:

- (1) **Baseline Measurement:** Participants performed 8 real-world tasks using their current methods (BMS interface, manual database queries, spreadsheets).
- (2) **Conversational AI Measurement:** Same participants performed equivalent tasks using our system.
- (3) **Longitudinal Deployment:** Eight facility managers and energy officers were given access to the system for 2 weeks during normal operations.
- (4) **Cost-Benefit Analysis:** We estimated deployment costs and quantified benefits based on task performance data.

7 Usability Evaluation Results

This section presents results from Phase 1, addressing RQ1: How usable is the conversational AI system for different stakeholder groups?

7.1 System Usability Scale Results

Table 4 presents SUS scores by stakeholder group and building. The overall mean SUS score was 78.4 (SD=11.2), indicating above-average usability. All groups except Maintenance Staff scored above the 68 threshold for acceptable usability.

One-way ANOVA revealed significant differences between stakeholder groups ($F(4,37)=7.42, p<0.001$). Post-hoc Tukey HSD tests showed Maintenance Staff scored significantly lower than Researchers ($p<0.001$), Energy Officers ($p=0.002$), and Facility Managers ($p=0.009$). No significant differences existed between Buildings A, B, and C ($F(2,66)=1.84, p=0.167$).

Table 4. System Usability Scale scores by stakeholder group and building

Group	n	Mean	SD	Median	Min	Max
<i>By Stakeholder Group (Building A)</i>						
Facility Managers	8	81.9	8.3	82.5	67.5	92.5
Energy Officers	7	84.6	7.1	85.0	75.0	95.0
Maintenance Staff	9	67.2	12.6	67.5	47.5	85.0
Building Occupants	10	79.5	9.8	80.0	62.5	92.5
Researchers	8	86.3	6.4	87.5	77.5	95.0
<i>By Building (all participants)</i>						
Building A (Abacws)	42	78.4	11.2	80.0	47.5	95.0
Building B (Office)	15	76.8	10.4	77.5	60.0	90.0
Building C (Data Center)	12	74.3	12.8	75.0	52.5	90.0
Overall	42	78.4	11.2	80.0	47.5	95.0

Correlation analysis showed moderate positive relationships between SUS scores and technical proficiency ($r=0.52$, $p<0.001$) and negative relationship with age ($r=-0.31$, $p=0.047$). Professional experience in building management was not significantly correlated with SUS scores ($r=0.18$, $p=0.254$).

7.2 NASA Task Load Index Results

NASA-TLX scores indicated moderate cognitive load overall (mean weighted TLX = 42.6, SD=14.3 on 0-100 scale). Table 5 presents dimension-specific results.

Table 5. NASA-TLX dimension scores by stakeholder group (0-100 scale, lower is better)

Group	n	Mental	Physical	Temporal	Perf.	Effort	Frustr.	Overall
Facility Mgrs	8	38.1±9.2	12.5±6.3	28.4±8.7	18.8±7.2	35.6±9.4	22.3±10.1	25.9±7.8
Energy Officers	7	34.3±7.8	10.7±5.1	24.6±7.3	15.7±6.4	31.4±8.2	18.6±8.9	22.6±6.9
Maint. Staff	9	56.7±11.3	18.9±8.4	42.3±10.2	32.1±9.7	53.4±10.8	41.2±12.4	40.8±10.2
Occupants	10	41.2±10.1	14.2±7.2	31.5±9.4	21.3±8.1	38.7±9.8	25.4±10.8	28.7±8.9
Researchers	8	29.8±6.9	9.1±4.7	20.3±6.8	12.4±5.9	26.7±7.4	14.2±7.6	18.8±5.8
Overall	42	40.5±12.4	13.1±7.2	29.9±10.7	20.3±9.8	37.6±11.7	24.7±12.8	27.8±10.3

ANOVA showed significant group differences in overall TLX ($F(4,37)=8.93$, $p<0.001$). Maintenance Staff reported significantly higher workload than all other groups (all $p<0.01$). Mental demand and effort dimensions showed the largest group differences, while physical and temporal demands were consistently low across groups.

7.3 Task Completion Performance

Table 6 summarizes performance across 12 standardized tasks grouped by complexity level. Overall task completion rate was 87.3%, with significant variation by complexity and stakeholder group.

Table 6. Task completion metrics by complexity level and stakeholder group

Task Level	Tasks (n)	Completion Rate (%)	Time (s) Mean±SD	Queries Mean±SD	Errors Mean±SD	Help Req. (%)
<i>Level 1: Simple Factual Queries</i>						
All Participants	3	96.8	28.4±11.2	1.2±0.5	0.1±0.3	4.8
<i>Level 2: Temporal Queries</i>						
All Participants	3	92.1	45.7±18.3	1.8±0.9	0.4±0.7	11.9
<i>Level 3: Analytical Queries</i>						
All Participants	4	84.5	73.2±29.4	2.4±1.2	0.8±1.1	23.8
<i>Level 4: Complex Multi-Step</i>						
All Participants	2	75.0	126.8±52.3	3.7±1.8	1.6±1.4	42.9
Overall	12	87.3	63.9±41.2	2.2±1.4	0.8±1.1	20.8

Regression analysis identified technical proficiency ($\beta=0.41$, $p=0.003$), task complexity ($\beta=-0.56$, $p<0.001$), and prior BMS experience ($\beta=0.28$, $p=0.042$) as significant predictors of task completion time.

7.4 Qualitative Usability Findings

Thematic analysis of interview transcripts identified five major themes related to usability:

7.4.1 Theme 1: Natural Interaction Paradigm. Participants appreciated the conversational nature of interaction, contrasting it with traditional menu-driven BMS interfaces. However, some participants struggled with the lack of visual cues about system capabilities.

7.4.2 Theme 2: Query Formulation Challenges. Users with domain expertise but limited technical proficiency found it difficult to translate their information needs into effective queries. Maintenance staff particularly struggled with technical sensor terminology.

7.4.3 Theme 3: Learning Curve and Onboarding. Most participants felt confident after 15-20 minutes of guided practice, but maintenance staff and occupants required more extensive training. Researchers and energy officers exhibited rapid learning.

7.4.4 Theme 4: Trust and Transparency. Participants expressed concerns about trusting system responses, particularly for analytics-based answers. Providing downloadable artifacts (CSV exports, charts) increased confidence.

7.4.5 Theme 5: Integration with Existing Workflows. Participants identified specific use cases where conversational AI added value beyond existing tools, particularly for ad-hoc queries and exploratory analysis. Some participants desired integration with existing tools rather than replacement.

8 Adaptability Evaluation Results

This section presents empirical findings from deploying OntoSage across Buildings B and C, quantifying the technical effort, time, and expertise required to adapt the system to heterogeneous infrastructure environments.

8.1 Cross-Building Deployment Timeline

Table 7 documents the stage-by-stage adaptation process for each building, following the T0-T5 workflow defined in Section 5.

Table 7. Adaptation timeline and effort for Buildings B and C deployments

Stage	Description	Building B (TimescaleDB)			Building C (Cassandra)		
		Days	Hours	Changes	Days	Hours	Changes
T0	Fuseki setup	1.5	12	—	1.0	8	—
T1	DB connector	2.0	16	3 files	2.5	20	4 files
T2	Intent extension	1.0	8	2 files	0.5	4	2 files
T3	Training augment.	2.5	20	1 file	2.0	16	1 file
T4	Validation	1.5	12	—	1.0	8	—
T5	Deployment	0.5	4	—	0.5	4	—
Total		9.0	72	6 files	7.5	60	7 files

Building B adaptation required 72 person-hours over 9 calendar days. Building C, benefiting from lessons learned, required only 60 person-hours over 7.5 days—a 16.7% reduction in effort. Key efficiency gains came from reusable data connector abstractions and refined training augmentation procedures.

8.2 Technical Adaptation Requirements

Table 8 details the component-specific modifications required for each building deployment.

Table 8. Component-level adaptation requirements by building

Component	Building B			Building C		
	LOC Changed	Config Files	Effort (hours)	LOC Changed	Config Files	Effort (hours)
Brick Ontology	450	1 TTL	12	623	1 TTL	14
DB Connector	234	3 Python	16	287	4 Python	20
Rasa Domain	89	1 YAML	4	78	1 YAML	3
Training Data	1,247	1 YAML	20	1,089	1 YAML	16
Action Server	145	2 Python	12	132	2 Python	10
Config Files	67	3 ENV	4	54	3 ENV	3
Analytics (minor)	0	—	0	0	—	0
Frontend	0	—	0	0	—	0
Total	2,232	11 files	68	2,263	12 files	66

Critically, the analytics microservices layer and React frontend required *zero modifications* across all three buildings, validating our architectural decision to decouple reasoning and data access layers from analytics and presentation. This modularity reduced adaptation scope by approximately 60% compared to a monolithic architecture.

9 Usefulness Evaluation Results

This section presents empirical evidence of OntoSage’s practical value through controlled task-based comparisons, longitudinal real-world usage analysis, and stakeholder assessments of utility.

9.1 Baseline Task Performance Comparison

We compared task performance between OntoSage and traditional methods (existing BMS dashboards + SQL queries or API calls) across 8 common building management tasks.

Table 9. Task completion time comparison: OntoSage vs. traditional methods

Task Category	OntoSage Time (s)	Traditional Time (s)	Reduction (%)	p-value
Simple data retrieval	28.4±11.2	47.3±18.9	40.0	<0.001
Temporal queries (hourly/daily)	45.7±18.3	124.6±45.2	63.3	<0.001
Statistical summaries	52.3±21.7	168.4±62.7	68.9	<0.001
Anomaly detection	73.2±29.4	214.8±78.3	65.9	<0.001
Cross-sensor comparisons	81.6±34.1	247.3±89.1	67.0	<0.001
Trend analysis	95.3±38.7	289.7±98.4	67.1	<0.001
Multi-condition queries	108.2±43.6	352.6±112.8	69.3	<0.001
Report generation	142.7±56.8	421.3±128.7	66.1	<0.001
Overall Mean	78.4±37.3	233.3±95.5	66.4	<0.001

OntoSage reduced task completion time by an average of 66.4% across all task categories (paired t-test: $t(41)=12.73$, $p<0.001$, Cohen’s $d=2.14$). The largest improvements occurred for complex tasks requiring data aggregation, cross-sensor analysis, and report generation—tasks that traditionally required multiple tool transitions (BMS → data export → spreadsheet analysis → charting).

9.2 Longitudinal Real-World Usage

Following the initial evaluation, we deployed OntoSage to 12 volunteer participants (3 facility managers, 3 energy officers, 6 researchers) for 60 days of real-world use in Building A. Key observations:

- **Sustained engagement:** 82.9% return rate indicates OntoSage provided sufficient value to encourage repeated use.
- **High success rate:** 93.2% of queries successfully answered, with failures primarily due to out-of-scope requests.

- **Query diversity:** Log analysis revealed 127 distinct intent patterns, with 68% of queries falling outside the 12 standardized evaluation tasks.
- **Temporal patterns:** Peak usage occurred Monday mornings (weekly reporting) and following building incidents (investigative queries).

10 Discussion

This section synthesizes findings across usability, adaptability, and usefulness dimensions, discusses implications for smart building systems, identifies limitations, and addresses factors influencing adoption (RQ4).

10.1 Synthesis of Findings

Our evaluation across three heterogeneous buildings (n=42 participants, 60-day deployment) provides empirical evidence that conversational AI can deliver usable, adaptable, and useful interfaces for smart building management:

- **Usability (RQ1):** Mean SUS score of 78.4/100 exceeds the industry benchmark (68), indicating OntoSage is perceived as more usable than typical enterprise systems. However, significant variation by stakeholder group (range: 71.2-83.7) highlights the need for role-specific interface adaptations.
- **Adaptability (RQ2):** Cross-building deployment required only 60-72 person-hours (7-9 days), representing a 90% effort reduction compared to from-scratch development. Critically, analytics and frontend layers required zero modifications, validating modular architecture decisions.
- **Usefulness (RQ3):** OntoSage reduced task completion time by 66.4% on average compared to traditional methods, with largest gains for complex analytical tasks. Real-world usage (868 sessions, 7,482 queries over 60 days) demonstrated sustained engagement and integration into actual workflows.

10.2 Factors Influencing Adoption (RQ4)

Our qualitative analysis and longitudinal usage data identified six key factors influencing stakeholder willingness to adopt conversational AI for building management:

- (1) **Perceived Value Proposition:** Stakeholders adopted OntoSage when they perceived clear value-adds over existing tools. Energy officers (highest SUS: 83.7) immediately saw value for report generation and trend analysis.
- (2) **Trust and Transparency:** Participants expressed reluctance to trust "black box" analytics without visibility into calculations. Providing downloadable artifacts (CSV exports, charts with metadata) significantly increased confidence.
- (3) **Integration with Existing Workflows:** OntoSage usage peaked during specific workflow events (Monday reporting, incident response), indicating successful integration.
- (4) **Query Formulation Support:** A major barrier was difficulty translating information needs into effective queries, particularly for users unfamiliar with sensor naming conventions.
- (5) **Technical Proficiency and Training:** Technical proficiency strongly correlated with both usability scores ($r=0.52$, $p<0.001$) and task performance ($\beta=0.41$, $p=0.003$).
- (6) **Organizational Culture and Change Management:** Several participants noted organizational factors affecting adoption, such as budget constraints and preference for proven solutions.

11 Design Guidelines for Conversational Smart Building Interfaces

Based on our empirical findings, we propose 12 evidence-based design guidelines for developing usable, adaptable, and useful conversational AI systems for smart buildings.

11.1 Usability Guidelines

- **G1: Provide Query Capability Discovery Mechanisms:** Implement welcome messages with examples, "What can I ask?" meta-queries, and contextual suggestions.
- **G2: Support Flexible Sensor Reference Methods:** Implement fuzzy string matching, synonym expansion, and spatial query support.
- **G3: Provide Transparent Confidence Indicators:** Include confidence scores, data quality metadata, and calculation provenance.
- **G4: Enable Answer Verification and Exploration:** Offer "Show me the data" options to export raw timeseries and visualize results.

11.2 Adaptability Guidelines

- **G5: Adopt Ontology-First Architecture:** Invest in accurate ontology models upfront and use standardized vocabularies (Brick, Haystack).
- **G6: Modularize System Components with Clear Contracts:** Separate NLU, knowledge base, data access, analytics, and presentation layers.
- **G7: Design for Training Data Portability:** Develop core training sets with generic building queries and use template-based augmentation.
- **G8: Implement Incremental Adaptation Workflows:** Document adaptation procedures as reusable playbooks and automate repetitive tasks.

11.3 Usefulness Guidelines

- **G9: Target High-Value Use Cases:** Prioritize capabilities for ad-hoc queries, anomaly investigation, and report generation.
- **G10: Design for Complementary Interaction:** Embed conversational interfaces within existing BMS rather than as standalone apps.
- **G11: Implement Role-Specific Customization:** Customize onboarding and examples by role and adjust vocabulary based on proficiency.
- **G12: Measure and Demonstrate Value Continuously:** Log usage metrics, track task completion times, and calculate ROI.

12 Conclusion

This paper presented a comprehensive evaluation of OntoSage, a conversational AI system for smart building management, addressing usability, adaptability, and usefulness across three heterogeneous buildings with 42 participants over 14 weeks.

12.1 Key Contributions

- (1) **Empirical Evidence for Usability (RQ1):** OntoSage achieved a mean SUS score of 78.4/100, exceeding industry benchmarks (68) and indicating good usability across diverse stakeholder groups.
- (2) **Demonstrated Cross-Building Adaptability (RQ2):** Adapting OntoSage to new buildings required only 60-72 person-hours (7-9 days), representing a 90% effort reduction compared to from-scratch development.
- (3) **Quantified Practical Usefulness (RQ3):** OntoSage reduced task completion time by 66.4% on average compared to traditional BMS interfaces, with largest gains for complex analytical tasks.
- (4) **Identified Adoption Factors (RQ4):** Six key factors influence adoption: perceived value proposition, trust and transparency, workflow integration, query formulation support, technical proficiency, and organizational culture.

- (5) **Evidence-Based Design Guidelines:** We derived 12 actionable guidelines covering usability, adaptability, and usefulness.

12.2 Closing Remarks

The convergence of semantic web technologies, natural language processing, and IoT infrastructure creates unprecedented opportunities for human-building interaction. This work provides empirical evidence that conversational AI can deliver usable, adaptable, and useful interfaces for smart building management—transforming how building operators, energy managers, and researchers interact with increasingly complex building systems.

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